



Instruction Manual for Electrical Technology Laboratory **Department of Electrical Engineering, IIT (ISM) Dhanbad**

Experiment No:01

VERIFICATION OF SUPERPOSITION THEOREM



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Experiment No.

TITLE: Verification of Superposition Theorem

OBJECTIVE: To verify the Superposition theorem experimentally.

APPARATUS USED:

Serial No.	Name of the Equipment	Rating	Model No. of the Equipment	Makers Name
1.				
2.				
3.				
4.				

THEORY:

Statement: If any network made up of linear resistances and containing more than one source (voltage source or current source), the resultant current flowing in any branch is the algebraic sum of currents that would flow in that branch if each source was considered separately, all other sources being replaced at that time by their respective internal resistances. In other words, a single source is considered at a time while other sources remain inactive.

How to inactivate a source: Replace ideal voltage source by short circuit and ideal current source by open circuit.

CIRCUIT DIAGRAM:

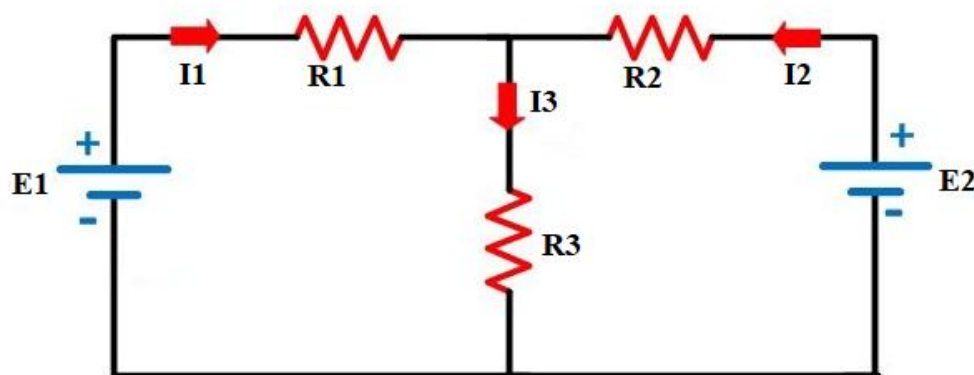


Fig: Circuit diagram for experimental setup

PROCEDURE:**For verification of Superposition Theorem**

- 1) Construct the circuit as shown in Fig.
- 2) Switch on the power supply of the module.
- 3) Keep E_1 active and E_2 inactive, measure the current through R_3 and name it as I_3' .
- 4) Keep E_1 inactive and E_2 active, measure the current through R_3 and name it as I_3'' .
- 5) Keep both the sources (E_1 and E_2) active and measure the current through R_3 and name it as I_3 .
- 6) Find I_{3S} as $I_{3S} = I_3' + I_3''$.
- 7) Compare I_3 and I_{3S} and find the percentage error.

CALCULATION:

$$\% \text{ Error} = \left[\frac{I_3 - I_{3S}}{I_3} \right] \times 100$$

COMMENTS/ DISCUSSIONS:

Write your comments on the results obtained and discuss the discrepancies, if any.

PRECAUTIONS:

- 1) Turn off the power switch to equipment before making connections.
- 2) Don't use broken connecting wires.
- 3) Maintain a work space clear of extraneous material such as books, papers, and clothes.
- 4) Don't wear loose clothes.
- 5) Wear shoes that cover the feet.

REPORT:

- 1) State the difference between linear and non-linear circuits. Give some examples of linear and non-linear elements in circuit theory.
- 2) State the difference between unilateral and bilateral circuits. Give some examples of unilateral and bilateral elements in circuit theory.
- 3) What are the limitations of Superposition theorem?
- 4) Can we apply Superposition theorem in circuit having non-linear elements?

DATA SHEET

Experiment No. _____

Date: _____

TITLE: Verification of Superposition Theorem.

Name: _____ Sem: _____ Year: _____

Adm. No. _____

Experimental Data:-

	E₁ (active)	E₂ (active)	E₁&E₂ (active)		Remarks
Sl. No.	I₃' (A)	I₃'' (A)	I₃ (A)	I_{3S}(A)=I₃' + I₃''	Percentage Error
1.					
2.					
3.					

(Signature of the teacher)